

SF-01: Lean verification of Newton H-matrix preservation

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Scope and attribution. This note records formalization of the complete original SF-01 target. Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge, retains mathematical authorship of the original resolution. His manuscript, *Preservation of positive-diagonal H-matrices by scaled Newton and Halley square-root iterations*, Theorem 1, remains unchanged. Sidney Holden retains authorship and Apache-2.0 credit for the two unchanged IV-03 Lean files genuinely imported by this proof.

Original target and exact formal theorem

For a real square matrix Y , its comparison matrix has entries

$$\mathcal{M}(Y)_{ii} = |Y_{ii}|, \quad \mathcal{M}(Y)_{ij} = -|Y_{ij}| \quad (i \neq j).$$

It is a nonsingular H-matrix if $\mathcal{M}(Y) = sI - B$ for an entrywise nonnegative real matrix B with $s > \rho(B)$, where ρ is the full complex algebraic spectral radius. For every $n \geq 1$ and every nonsingular real H-matrix $A \in \mathbb{R}^{n \times n}$ with strictly positive diagonal, define

$$X_0 = A, \quad X_{k+1} = \frac{1}{2}(X_k + X_k^{-1}A) \quad (k \geq 0).$$

The Lean theorem `NLA.SF01.canonical_newton_preservation` proves that every X_k is invertible and a real nonsingular H-matrix with strictly positive diagonal. Every positive dimension and every iteration is included. No symmetry, normality, entrywise positivity, finite iteration bound or assumed rational representation is imposed.

The proof establishes both directions between the original spectral condition and positive diagonal-dominance weights. Actual shifted resolvent domination and block inverse identities produce a finite rational invariant for every Newton step. Empty pole families, repeated poles and zero weights are included. The manuscript's scaled or affine initializations and Halley extension are outside this formal claim; they are not needed for the original target.

Executed verification and reproduction

The immutable proof revision is

3312b0795873cfecade03fa421a5433651d47674.

Actual non-root Linux run 35175258802, job 105055517720, checked all 271 submitted inputs and all 24 independent Comparator contracts. Default-kernel replay, Comparator, transitive axiom checks and the required rejection, regression and sandbox controls passed. All exports use only `propext`, `Classical.choice` and `Quot.sound`. The deliberate Challenge placeholders are specifications and are never imported by the implementation.

Lean 4.33.1, Mathlib revision 0df444a360eaa60ab8c11dca51a86af692955474, and LeanCert revision 621a43d7cf21f87872392a01e874f2f1dbddc926 are pinned. The sole interval certificate proves $0 < 1/2$ with LeanCert kernel trust and is consumed by initialization and coefficient halving. All variable matrix, spectral and iteration arguments are symbolic; no interval subdivision or numerical sampling is used.

With the pinned dependencies, run `lake build` in

`matrix-functions-and-stability/SF-01/lean`.

The project README gives the full shared Linux bootstrap, self-test and verification commands, source correspondence, logs and axiom report. The final Solution import closure has 31 files; all 37 selected sources, including six earlier export probes, are retained and distinguished.

Two complete nonauthor source reviews and a separate independent runtime evidence audit are retained with explicit reviewer roles. Substantial OpenAI Codex assistance is disclosed. Neither these agent reviews nor the runtime checks assert official Tau Ceti endorsement, human peer review or checker infallibility. Local serial macOS development is a distinct execution from the Linux run above. Auditing a recorded run is not another execution. Later publication-commit and upstream PR checks are recorded separately.